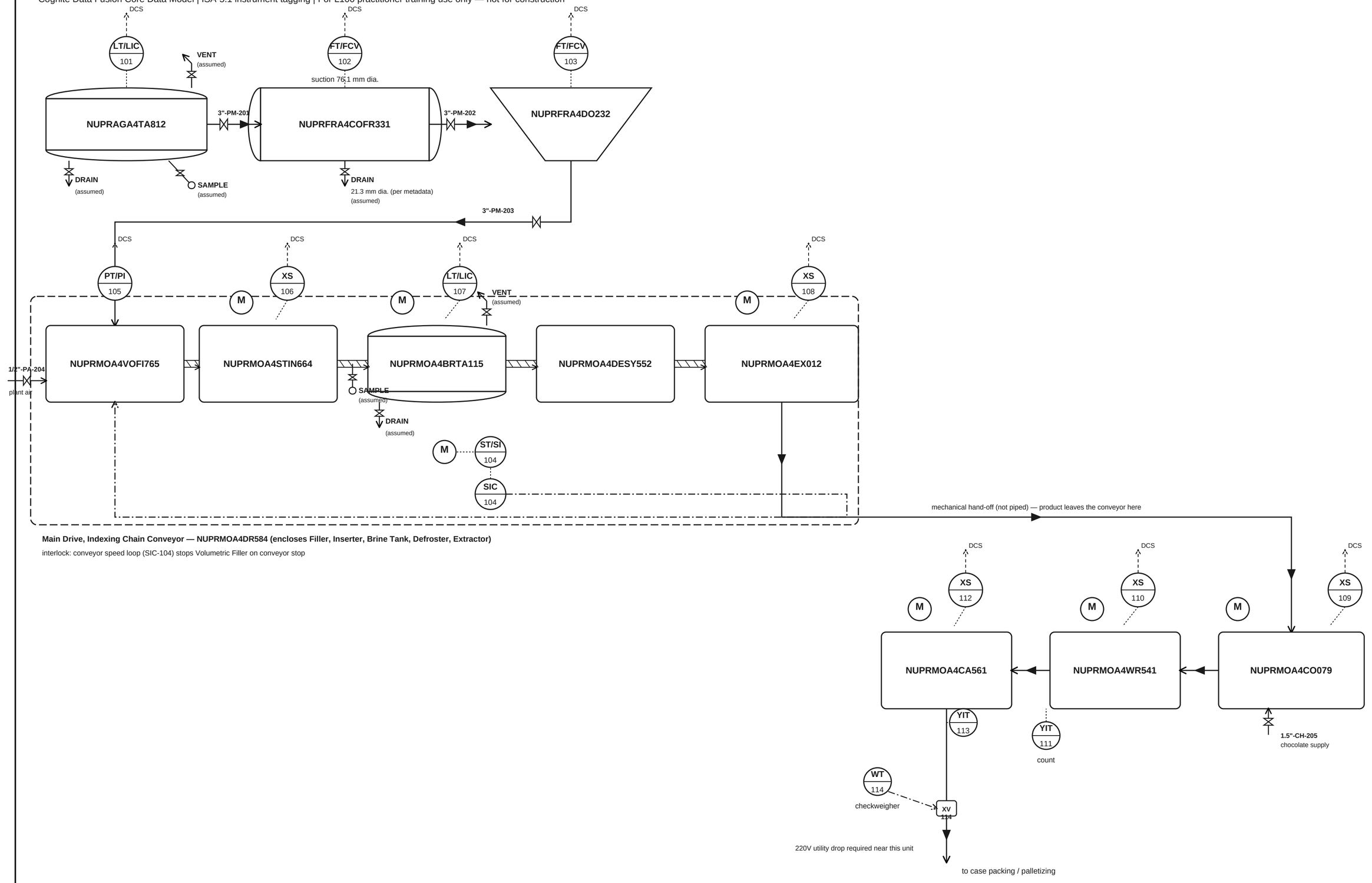
P&ID (PRELIMINARY) — NUREMBERG SITE — MOULDING LINE A46 — ICE CREAM FACTORY DATA

Cognite Data Fusion Core Data Model | ISA-5.1 instrument tagging | For L100 practitioner training use only — not for construction



Main Drive, Indexing Chain Conveyor — NUPRMOA4DR584 (encloses Filler, Inserter, Brine Tank, Defroster, Extractor)

interlock: conveyor speed loop (SIC-104) stops Volumetric Filler on conveyor stop

Legend

Solid line + ►	Process piping / product flow (arrowheads = assumed direction)
Bowtie (on a line)	Manual isolation valve
size"-CODE-###	Line number: size-service code-sequence (PM=mix, PA=plant air, CH=chocolate)
Short dashed line	Instrument signal to PLC / DCS
Dash-dot line	Safety interlock / control logic
Long dashed box	Equipment grouping enclosure (e.g. Main Drive conveyor)
Circle (M)	Electric motor / servo drive
Bowtie + flag	Vent (assumed location — open to atmosphere)
Bowtie + arrow down	Drain (assumed location)
Bowtie + circle	Sample point (assumed location)
LT / LIC	Level transmitter / level indicating controller (tanks)
FT / FCV	Flow transmitter / flow control valve (dosing, throughput)
PT / PI	Pressure transmitter / indicator (pneumatic supply)
ST / SI / SIC	Speed transmitter / indicator / speed indicating controller
XS / YI	Motor state switch / status indicator
YIT / YC	Event or pulse counter (count variable)
WT / AE	Checkweigher / vision inspection (quality variable)
XV / YV	Solenoid reject valve / pneumatic pusher

Notes / open items before this drawing is construction-ready

Line numbers and valve locations are illustrative — sizes, services, and sequence numbers are assumed, not taken from a line list or spec. Actual 76.1/21.3 mm freezer dia. is metadata.

Filler -> Stick Inserter -> ... -> Cartoning connectors carry NO line numbers/valves on purpose: that path is mechanical product transfer on the indexing chain conveyor, not piped flow.

Plant air (PA-204) and chocolate supply (CH-205) are inferred utility tie-ins, not present in the source data.

Vent, drain, and sample points (bowtie symbols) are shown using standard engineering assumptions — their locations are NOT confirmed against as-built piping or P&ID data.

planned_status (Core Data Model time series) is a software/ planning variable only — intentionally not drawn on this P&ID.

Stick Inserter needs physical safety guarding (fencing / light curtain) around moving parts — not yet specified in source data.

Brine Tank circulation is shared with the Brine Flow System (NUPRMOA4BRFLSY561) — modeled as one loop on this drawing.

Defroster System has no motor or instrument data in the source — its status point requires field verification.

Instrument loop numbers (101-114) are illustrative for this training exercise, not an assigned plant numbering series.

Source: Ice Cream Factory data, Nuremberg site, Moulding Line A46, modeled via the Core Data Model (ProductionLine / Unit / Equipment).

Drawing No.	NUR-P&ID-101
Client / project	Ice Cream Factory — L100 practitioner training
Site	Nuremberg
Unit	Moulding Line A46
Drawing type	P&ID — preliminary (ISA-5.1)
Data source	Core Data Model (ProductionLine, Unit, Equipment)
Status	For training use — not for construction
Rev.	0 — draft